

FinNews AI: Financial News, Made Simple

Project Documentation:

FinNews AI is a financial news simplification web application designed to help people understand markets, business, and economic news without the complicated financial jargon. The application brings the latest financial headlines, a market snapshot, and an AI-powered article simplifier into a single, lightweight website. Users can choose a story from the latest news, or paste an article or a news link, and receive a plain-English summary, key points, explanations of financial terms, and a short note on why the story matters.

The main objective of FinNews AI is to make financial reporting readable. Instead of searching for the meaning of unfamiliar terms while reading a headline, users get the meaning of the story itself. The application is designed as an educational tool: a way to learn how to read the news, not a replacement for it. It is available online at <https://aniketghodke2007-cmyk.github.io/FinNews-AI-Financial-News-Simplifier/>.

A major design principle of FinNews AI is that the AI simplifies and explains rather than advises. Headlines are retrieved from a news source, the article is rewritten in plain language by an AI model, and the result is presented together with clear signs of where it came from. Each story names its publisher, links back to the original article, and carries an AI-generated label. The application states that it is not financial advice and that AI-generated summaries may contain errors.

1. Scenarios

Scenario 1: The first scenario demonstrates how a user discovers financial news on FinNews AI. The user opens the home page, which presents the message "Financial News, Made Simple." along with a short explanation that the application helps users understand markets, business, and economic news without complicated jargon. Beside the message, a sample card shows the idea in action. The original headline, "Federal Reserve maintains restrictive monetary policy amid persistent inflationary pressures," is shown next to its AI-generated simplified version: "The US central bank is keeping interest rates high because inflation is still a concern."

The user then scrolls to the Latest Financial News section, which invites them to pick a story and simplify it in one click. Category buttons (All, Markets, Stocks, Economy, Business, and Technology) help narrow the list. Each story card shows its category, headline, a short description, the publisher, how long ago it was published, and a Simplify button. For example, the demonstration stories include "Fed holds rates steady, signals patience as inflation stays above target" (Economy), "Global equities edge higher as investors weigh earnings and bond yields" (Markets), and "Chipmaker shares climb after upbeat guidance on data-centre demand" (Stocks).

Activities

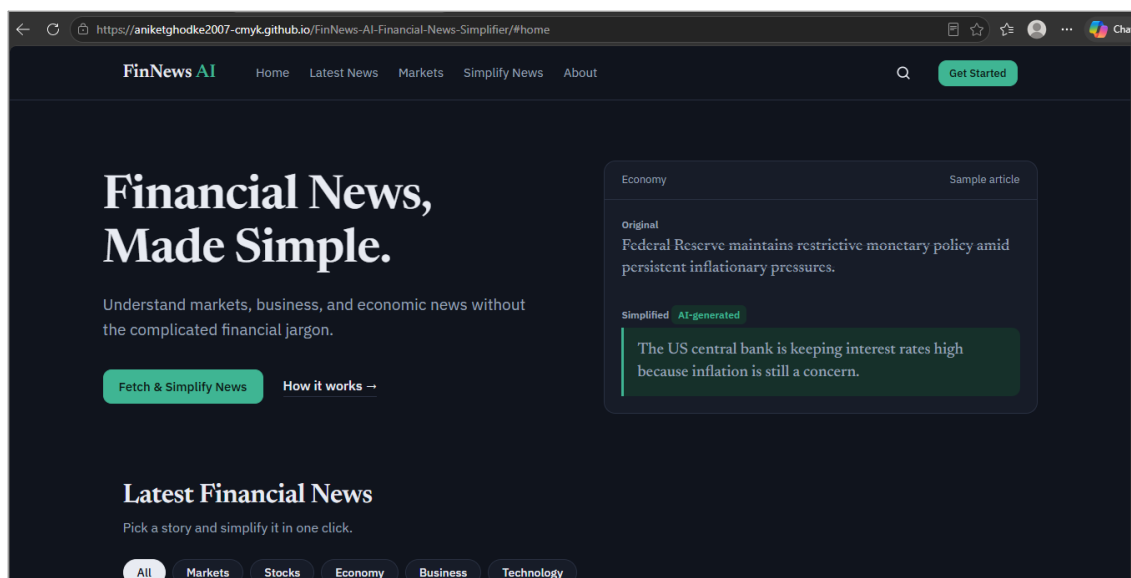
Activity 1.1: The user opens FinNews AI and reads the headline and the short introduction on the home page.

Activity 1.2: The user studies the sample card that compares an original headline with its simplified version.

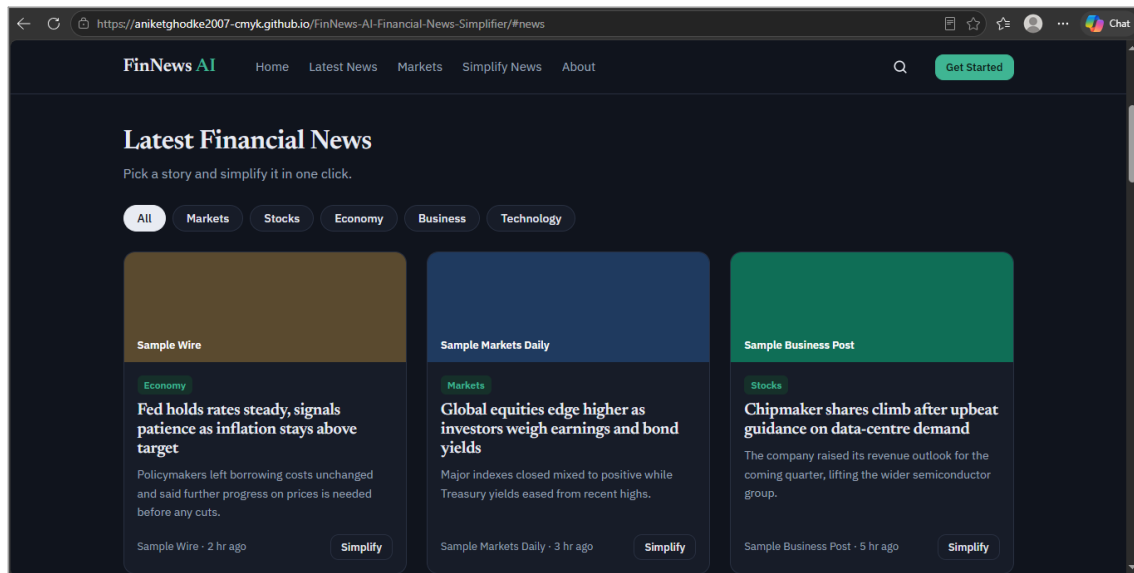
Activity 1.3: The user scrolls to Latest Financial News and selects a category such as Economy or Markets.

Activity 1.4: The application displays story cards with category, headline, description, publisher, and time.

Activity 1.5: The user clicks Simplify on a story to send it for simplification.



Description: The home page shows the FinNews AI logo and navigation bar (Home, Latest News, Markets, Simplify News, About) with a Get Started button. The hero section carries the headline "Financial News, Made Simple." and two actions, Fetch & Simplify News and How it works. On the right, a sample Economy card shows the original headline and the simplified version marked with an AI-generated badge.



Description: The Latest Financial News section shows the category filter buttons and a row of story cards. Each card displays a publisher banner, a category tag, the headline, a short description, the publisher with the time since publication, and a Simplify button. The stories shown use sample publishers such as Sample Wire, Sample Markets Daily, and Sample Business Post for demonstration.

Scenario 2: The second scenario demonstrates how users turn a difficult financial article into a clear explanation. In the Simplify News section, titled "Understand the Story in Seconds," the user can paste the full text of a financial news article or enter a link to one. The user can also choose a story from the Latest Financial News list instead of pasting anything.

When the user clicks Simplify Article, the request is sent to the FinNews AI backend, which asks the language model to rewrite the story in plain language. The result appears in the panel beside the input box and contains four parts: a plain-English summary, key points, explanations of financial terms, and why the story matters. Every summary is labelled as AI-generated, names its publisher, and links back to the original article, so the user can always check the full report.

Activities

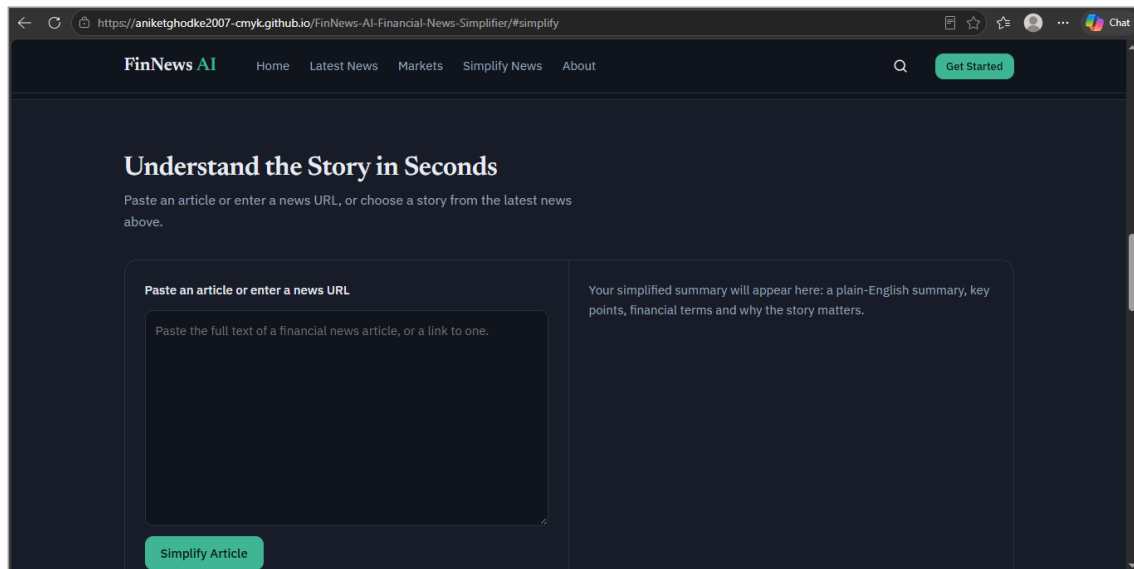
Activity 2.1: The user opens the Simplify News section from the navigation bar.

Activity 2.2: The user pastes an article or enters a news URL, or selects a story from the latest news.

Activity 2.3: The user clicks Simplify Article to start the simplification.

Activity 2.4: The backend passes the story to the AI model, which removes unnecessary financial jargon.

Activity 2.5: The application displays the summary, key points, financial terms, and why the story matters.



Description: The Simplify News section is split into two panels. The left panel contains the label "Paste an article or enter a news URL", a large text box, and the Simplify Article button. The right panel shows a message explaining that the simplified summary will appear there, made up of a plain-English summary, key points, financial terms, and why the story matters.

Scenario 3: The third scenario demonstrates the Market Snapshot and the trust features of FinNews AI. The Market Snapshot gives readers a quick view of major indices and instruments alongside the news. The section states clearly that it shows sample data for demonstration and that live prices will appear once a market data source is connected.

The application also explains how it works and who it is for, and includes a Trust & Transparency section. It states that FinNews AI simplifies financial information for educational and informational purposes, that summaries are AI-generated and should be checked against the original source, and that it explains the news without telling anyone what to buy or sell. The About section and the footer repeat this message with a disclaimer.

Activities

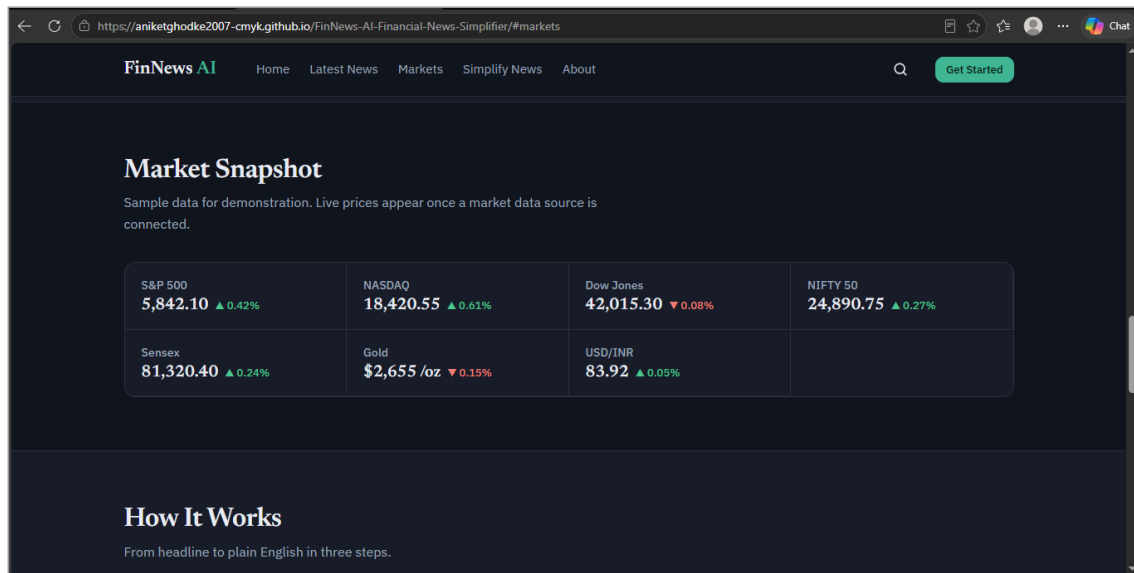
Activity 3.1: The user opens the Markets section to view the Market Snapshot.

Activity 3.2: The application shows each index or instrument with its value and percentage change, coloured green for gains and red for declines.

Activity 3.3: The user reads How It Works and Who It's For to understand the steps and the audience.

Activity 3.4: The user reviews Trust & Transparency: source attribution, original article link, AI-generated label, and the not-financial-advice statement.

Activity 3.5: The user reads the About section and the footer disclaimer to confirm how the summaries are produced and their limits.

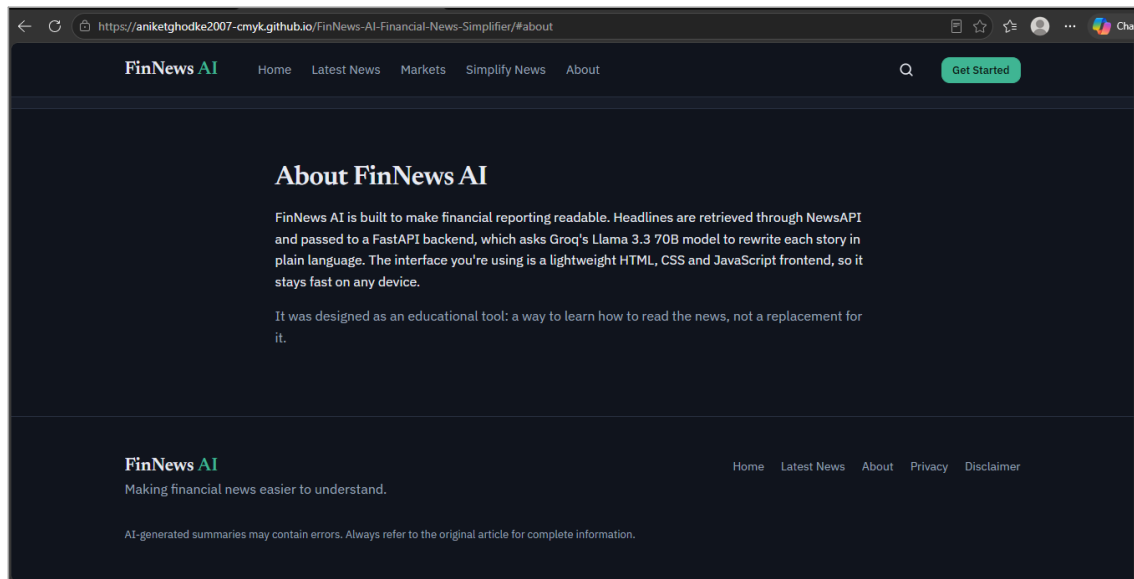


Description: The Market Snapshot shows a grid of market cards for S&P 500, NASDAQ, Dow Jones, NIFTY 50, Sensex, Gold, and USD/INR. Each card lists the value and the percentage change, with a green upward arrow for gains and a red downward arrow for declines. A note under the heading explains that the data is sample data for demonstration. The start of the How It Works section is visible below.

Example

The table below summarizes the sample values shown in the Market Snapshot of the demonstration site. These values are sample data, not live prices.

Market	Sample value	Change	Direction
S&P 500	5,842.10	0.42%	Up
NASDAQ	18,420.55	0.61%	Up
Dow Jones	42,015.30	0.08%	Down
NIFTY 50	24,890.75	0.27%	Up
Sensex	81,320.40	0.24%	Up
Gold	\$2,655 /oz	0.15%	Down
USD/INR	83.92	0.05%	Up



Description: The About FinNews AI section explains that headlines are retrieved through NewsAPI and passed to a FastAPI backend, which asks the Llama 3.3 70B model on Groq to rewrite each story in plain language, and that the interface is a lightweight HTML, CSS, and JavaScript frontend. It also says the tool was designed as an educational tool. The footer repeats the tagline "Making financial news easier to understand", lists links (Home, Latest News, About, Privacy, Disclaimer), and warns that AI-generated summaries may contain errors.

2. Technical Architecture

2.1 Architecture of FinNews AI

FinNews AI follows a simple, modular web architecture designed to retrieve financial news and present it in plain language. The architecture connects the user, a lightweight frontend, a news source, a backend service, and an AI language model. The system is designed so that the AI is used for one clear purpose, rewriting a story in simple language, while the source and original link stay visible to the user.

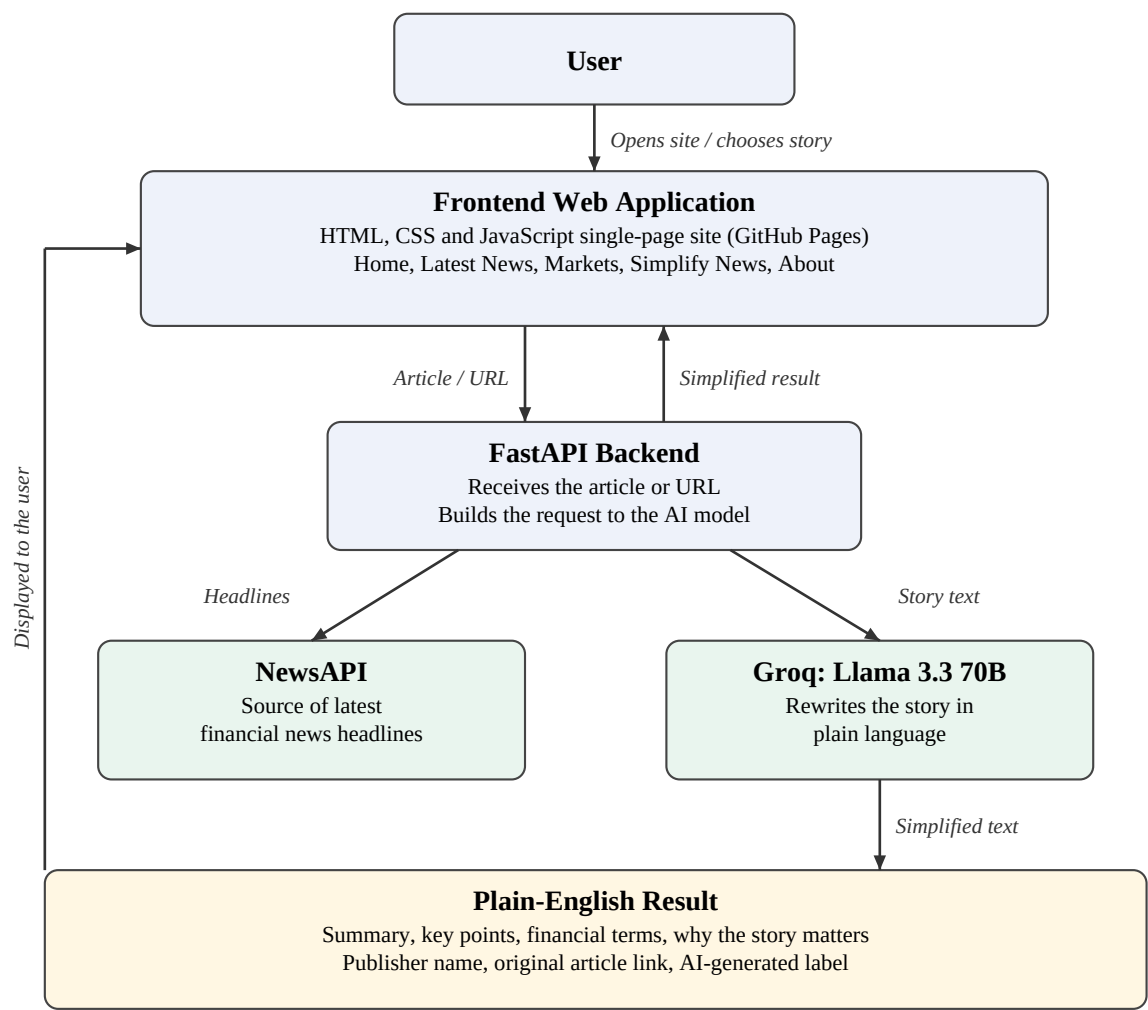


Figure 1: Technical Architecture of FinNews AI

2.2 Frontend Web Application

The frontend is the main interaction layer of FinNews AI. It is a lightweight site built with HTML, CSS, and JavaScript, hosted on GitHub Pages, so it stays fast on any device. Each section is reached through the navigation bar or an anchor in the address bar (for example, #home, #news, #markets, #simplify, and #about). A Get Started button leads directly to the simplifier. The interface uses cards, filter buttons, a two-panel simplifier, and a dark theme with green highlights.

2.3 News Retrieval

Headlines are retrieved through NewsAPI. The Latest Financial News section groups stories by category, namely Markets, Stocks, Economy, Business, and Technology, and shows the publisher and time of each story. The stories visible on the demonstration site use sample publishers.

2.4 FastAPI Backend

The FastAPI backend sits between the frontend and the AI model. It receives the article text or link, prepares the request for the language model, and returns the simplified result to the frontend. Keeping this logic in a backend service allows the frontend to remain small and fast.

2.5 AI Simplification

The backend asks Groq's Llama 3.3 70B model to rewrite each story in plain language. The model reads the article, removes unnecessary financial jargon, and produces a concise summary, key points, explanations of financial terms, and the context of why the story matters. The AI is used for explanation only; it is not used to give investment recommendations.

2.6 Simplified Output

The output panel shows four elements: a plain-English summary, key points, financial terms, and why the story matters. Each summary carries an AI-generated label and a timestamp, names its publisher, and includes a link to the original article.

2.7 Market Snapshot

The Market Snapshot displays values for S&P 500, NASDAQ, Dow Jones, NIFTY 50, Sensex, Gold, and USD/INR with percentage changes. The values are currently sample data for demonstration, and live prices will appear once a market data source is connected. Green and red colours are used consistently to show gains and declines.

2.8 Responsible Use

FinNews AI is designed as an educational tool. It states that summaries are AI-generated and should be checked against the original source, that it is not financial advice, and that AI-generated summaries may contain errors. The original article is always one click away.

2.9 Hosting and Deployment

FinNews AI is published as a web application on GitHub Pages, which makes it accessible from any modern browser through a public link without any installation by the user.

2.10 Architectural Principle

The central architectural principle of FinNews AI is:

News Source > FastAPI Backend > AI Simplification > Plain-English Result + Source Attribution

3. Project Workflow

The development of FinNews AI was organized into six major milestones. Each milestone focused on a specific stage of development, starting from requirement analysis and design and continuing through frontend development, news retrieval, AI integration, trust features, testing, and deployment.

Milestone 1: Requirement Analysis, Design Direction and Architecture

The first milestone focused on understanding the problem: financial news is often full of jargon that makes it hard for students, retail investors, and everyday readers to follow. The requirements included a news feed, an article simplifier, a market snapshot, and clear trust and transparency features. The architecture was planned around a lightweight frontend, a backend service, and an AI model. A dark visual theme with green accents was chosen for the interface.

Activities

Activity 1.1: Analyse the need for a tool that explains financial news in plain language.

Activity 1.2: Identify the major sections: Home, Latest News, Markets, Simplify News, and About.

Activity 1.3: Design the architecture connecting the frontend, news source, backend, and AI model.

Activity 1.4: Select the required tools: HTML, CSS, JavaScript, NewsAPI, FastAPI, and Groq's Llama 3.3 70B.

Milestone 2: Frontend Development

The second milestone built the user interface. The home page was created with the headline, the call-to-action buttons, and the sample Original versus Simplified card. The navigation bar, the Latest Financial News cards with category filters, and the Market Snapshot section were then developed with a responsive layout.

Activities

Activity 2.1: Develop the home page hero section and the sample simplification card.

Activity 2.2: Develop the navigation bar and the Get Started button.

Activity 2.3: Develop the news cards and category filter buttons.

Activity 2.4: Develop the Market Snapshot grid with colour-coded changes.

Activity 2.5: Improve responsiveness and usability across devices.

Milestone 3: News Retrieval and Backend

The third milestone connected the application to its data. Headlines are retrieved through NewsAPI and passed to a FastAPI backend, which handles the requests coming from the frontend.

Activities

Activity 3.1: Connect the application to NewsAPI to retrieve financial headlines.

Activity 3.2: Build the FastAPI backend to receive article text or links.

Activity 3.3: Connect the frontend to the backend so stories can be sent with one click.

Activity 3.4: Display publisher names and publication times on each story card.

Milestone 4: AI Simplification

The fourth milestone integrated the language model. The backend asks Groq's Llama 3.3 70B model to rewrite each story in plain language and to return a structured result. The simplifier interface was built so that the result appears beside the input.

Activities

Activity 4.1: Integrate the Llama 3.3 70B model through Groq with the backend.

Activity 4.2: Design the request so the model returns a summary, key points, financial terms, and why the story matters.

Activity 4.3: Develop the Simplify News section with the input box and result panel.

Activity 4.4: Display the simplified result with an AI-generated label.

Milestone 5: Trust, Transparency and Testing

The fifth milestone focused on responsible use and quality. Source attribution, the original article link, the AI-generated label, and the not-financial-advice statement were added. The main workflows were tested to check that stories can be selected, simplified, and displayed correctly.

Activities

Activity 5.1: Add source attribution and the original article link to every story.

Activity 5.2: Add the AI-generated label and timestamp to every summary.

Activity 5.3: Add the Trust & Transparency section, the About section, and the footer disclaimer.

Activity 5.4: Test selecting a story, pasting an article, and viewing the simplified result.

Activity 5.5: Test the layout across screen sizes.

Milestone 6: Documentation, Deployment and Conclusion

The sixth milestone focused on final review, deployment, and documentation. The application was published on GitHub Pages so that it can be accessed online, and screenshots of the important sections were captured for the project documentation.

Activities

Activity 6.1: Perform final functional testing of the application.

Activity 6.2: Deploy the application on GitHub Pages.

Activity 6.3: Capture screenshots of the important sections.

Activity 6.4: Prepare the project scenarios.

Activity 6.5: Prepare the technical architecture and six-milestone workflow documentation.

4. Website Pages / Modules

4.1 Home

The Home section introduces the application with the message "Financial News, Made Simple." and the line "Understand markets, business, and economic news without the complicated financial jargon." Two actions are provided: Fetch & Simplify News, which leads to the simplifier, and How it works, which leads to the explanation of the steps. A sample card shows an original headline beside its AI-generated simplified version.

4.2 Latest News

The Latest News section lists current financial stories as cards. Users can filter by All, Markets, Stocks, Economy, Business, or Technology. Each card shows the category, headline, short description, publisher, time, and a Simplify button that sends the story to the simplifier in one click.

4.3 Markets

The Markets section shows the Market Snapshot for S&P 500, NASDAQ, Dow Jones, NIFTY 50, Sensex, Gold, and USD/INR. Each item shows its value and percentage change with green and red indicators. The section is marked as sample data for demonstration until a live market data source is connected.

4.4 Simplify News

The Simplify News section, titled "Understand the Story in Seconds," lets users paste an article or enter a news URL, or choose a story from the latest news. After the user clicks Simplify Article, the result panel shows a plain-English summary, key points, financial terms, and why the story matters.

4.5 How It Works and Who It's For

How It Works explains the process in three steps: Find the news (latest financial news is collected from trusted sources), AI simplifies it (the language model removes unnecessary financial jargon), and Understand the story (the user gets a concise summary, key points, term explanations, and context). Who It's For describes the four groups of users the application serves.

Audience	How FinNews AI helps
Retail investors	Quickly understand market-moving news.
Students	Learn difficult financial concepts through simple explanations.
Professionals	Get important economic developments before meetings.
Everyday readers	Understand business and economic news without financial expertise.

4.6 Trust & Transparency

This section explains how FinNews AI keeps its summaries accountable. Every story names its publisher, each summary links back to the original article, every summary carries an AI-generated label and timestamp, and the application states that it is not financial advice: it explains the news but does not tell users what to buy or sell.

4.7 About

The About section describes how the application works: headlines are retrieved through NewsAPI and passed to a FastAPI backend, which asks Groq's Llama 3.3 70B model to rewrite each story in plain language, and the interface is a lightweight HTML, CSS, and JavaScript frontend. It explains that the tool was designed as an educational tool, and the footer adds links to Privacy and Disclaimer pages along with a warning that AI-generated summaries may contain errors.

5. Conclusion

FinNews AI successfully combines financial news with AI-assisted simplification in a single lightweight web application. The application provides a feed of the latest financial stories, a market snapshot, and a simplifier that turns complicated articles into a plain-English summary, key points, term explanations, and context.

The project places particular emphasis on transparency and responsible use. Every story names its publisher and links to the original article, every summary is labelled as AI-generated, and the application clearly states that it is an educational tool and not financial advice.

The development process was completed through six milestones covering requirement analysis and architecture, frontend development, news retrieval and backend, AI simplification, trust and testing, and final documentation and deployment.

Overall, FinNews AI demonstrates how a modern web application can use a news source, a backend service, and a large language model to make financial reporting readable for students, retail investors, professionals, and everyday readers.